

SADMAN SHAWRAZ

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RESEARCH & TECHNICAL SUMMARY

4+ years of research experience in antibody discovery, protein engineering, B-cell immunology, and functional assays. Experienced in synthetic VHH library design, ribosome display, mammalian cell-based selection, targeted diversification, NGS analysis, protein binding, pseudotyped virus neutralization, and mouse B-cell phenotyping. Strong background in developing and troubleshooting complex experimental workflows from library design through functional validation.

CORE SKILLS

Antibody Discovery: VHH/Nanobody Engineering • Synthetic Library Design • Ribosome Display • Cell-Based Selection • Affinity Maturation

Cell & Functional Assays: Mammalian Cell Engineering • Flow Cytometry • Primary Cell Phenotyping • ELISA • Pseudotyped Virus Neutralization • Mouse Models

Molecular & Protein Biology: Protein Expression & Purification • Protein Binding Assays • RT-qPCR • NGS

RESEARCH EXPERIENCE

Research Assistant, Bieniasz Lab, Rockefeller University, New York, NY.

September 2024-Present

- Developing a synthetic, animal-free VHH/nanobody discovery and maturation platform integrating in vitro selection, mammalian cell-based selection, targeted diversification, and functional characterization.
- Designed and constructed highly diverse synthetic VHH libraries and enriched target-specific binders through ribosome display and affinity pulldown.
- Engineered Ba/F3 and DHFR-knockout 293T cell-based selection systems that couple VHH-antigen recognition to survival.
- Mimicked affinity maturation by introducing targeted VHH sequence variations using AID-T7-mediated mutagenesis.
- Tracked VHH diversity, enrichment, and mutation patterns across selection rounds using High Throughput Sequencing.
- Characterized antibody and nanobody function using luciferase-based SARS-CoV-2 pseudotyped virus neutralization, and protein-binding assays.

Research Technician, Genetics and Immunology, Vuong Lab, CCNY, New York, NY

May 2022-August 2024

- Investigated RHAU/DHX36 regulation of AID-dependent Class Switch Recombination (CSR) and somatic hypermutation using conditional RHAU knockout mice and ex vivo B-cell models.
- Characterized RHAU function by measuring serum immunoglobulin responses following immunization, ex vivo CSR in purified B cells, and germline switch transcripts by RT-qPCR.
- Performed immunophenotyping of human PBMCs to characterize NK-cell exhaustion and B- and T-cell populations in chronic HBV patients.

Research Assistant, Materials Research, RF CUNY, New York, NY

May 2021-December 2021

- Evaluated activated carbon performance from NYC wastewater odor-control vessels by testing H₂S adsorption and reporting efficacy data to DEP/plant teams.

EDUCATION

BSc, Biochemistry [Minor: Earth and Atmospheric Sciences], The City College of New York, June 2022

GPA: **3.73**

(Transferred from) BSc, Biotechnology, BRAC University, Dhaka, Bangladesh (2015-2016)

GPA: **3.99**

PUBLICATIONS

- Wiatr et al., 2026. Memory B cell development in response to mRNA SARS-CoV-2 and nanoparticle immunization in mice. **Shawraz, S.** co-author. Proceedings of the National Academy of Sciences.
- Deimel et al., 2025. Clonal expansion and diversification of germinal center and memory B cell responses to booster immunization in primates. **Shawraz, S.** co-author. Cell Reports.
- Vuong, B. (2024). RNA Helicase associated with AU rich element (RHAU) regulates class switch recombination (CSR) and gene expression in B Cells. **Shawraz, S.** co-author. Manuscript in writing.

TRAINING & PRESENTATIONS

Poster Presentation, Immunology2024, American Association of Immunologists (AAI), 2024

Postbaccalaureate Training in Biochemistry, Biophysics, and Bio-design (B3-RaMMMP), CCNY, 2023-24

HONORS & AWARDS

Magna cum laude, The City College of New York, 2022

Frank & Rose Brescia Award II, The City College of New York, 2021

Dean's List, CUNY City College, 2019 & 2020

Merit Scholarship based on Academic Results, BRAC University, 2016